

Generating Sketches From Face Images

Final Presentation

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Problem Statement

Generating sketches from face photographs by unsupervised learning

Our aim: improving (Chan et al., 2022)'s model on face images.

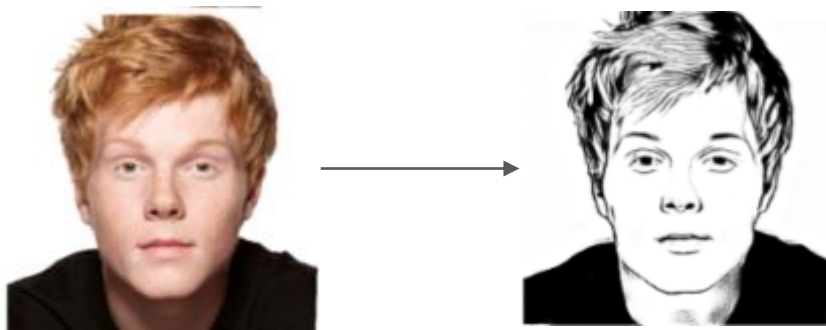


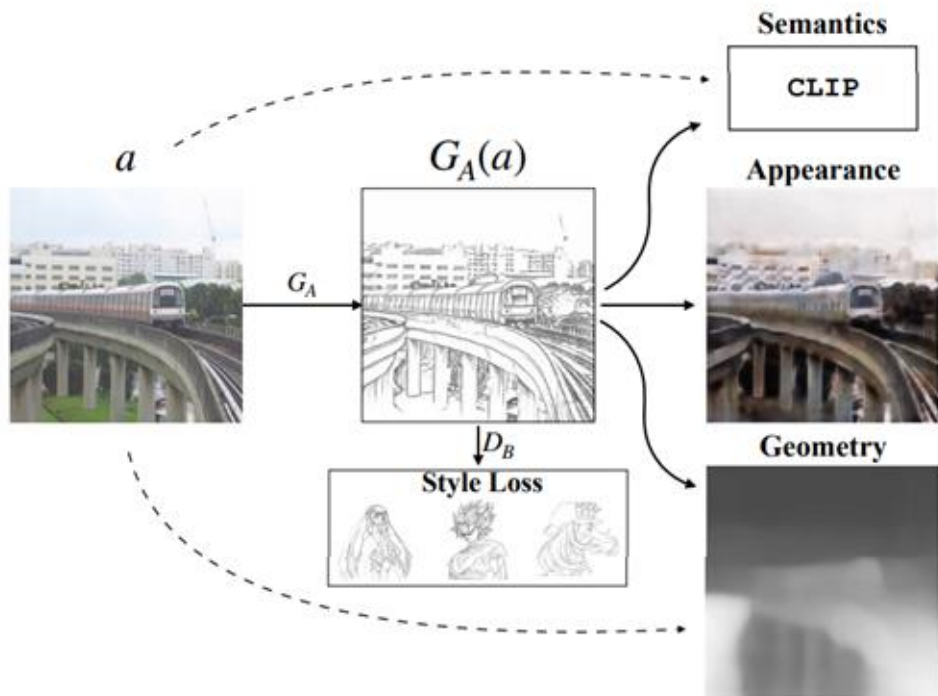
Figure Adapted from ([Chan et al. 2022](#))

Related Work

Some notable works on face sketch synthesis are:

- CA-GAN (Gao et al., 2017)
- ApdrawingGAN (Yi et al., 2019)
- UPDG (Yi et al., 2020)
- Chan et al. (2022)'s work

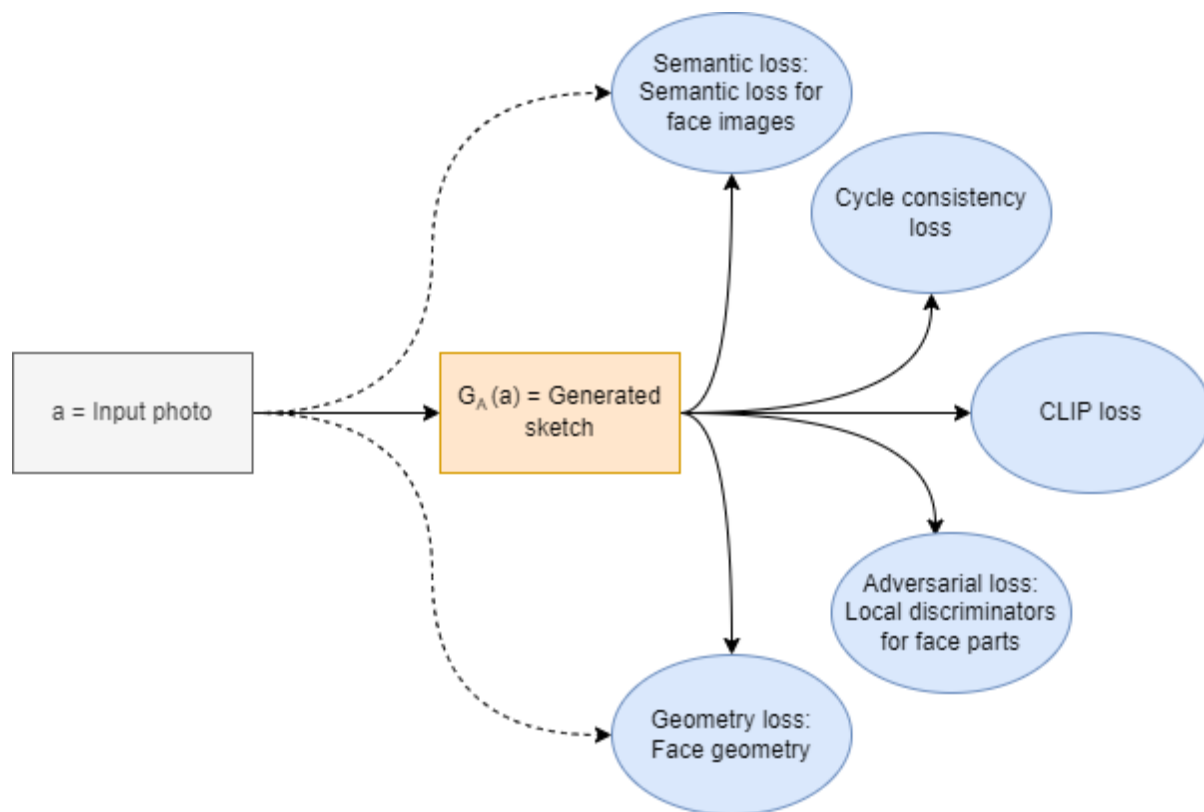
Baseline Overview



Baseline: “Learning to Generate Line Drawings that Convey Geometry and Semantics” (Chan et al., 2022)

- Style Loss: Adversarial Loss
- Semantics Loss: CLIP (Radford et al., 2021)
- Appearance: CycleGAN (Zhu et al., 2017)
- Geometry Objective: Depth Network (Wang et al., 2017)

Model overview



Generator and Cycle Consistency Loss

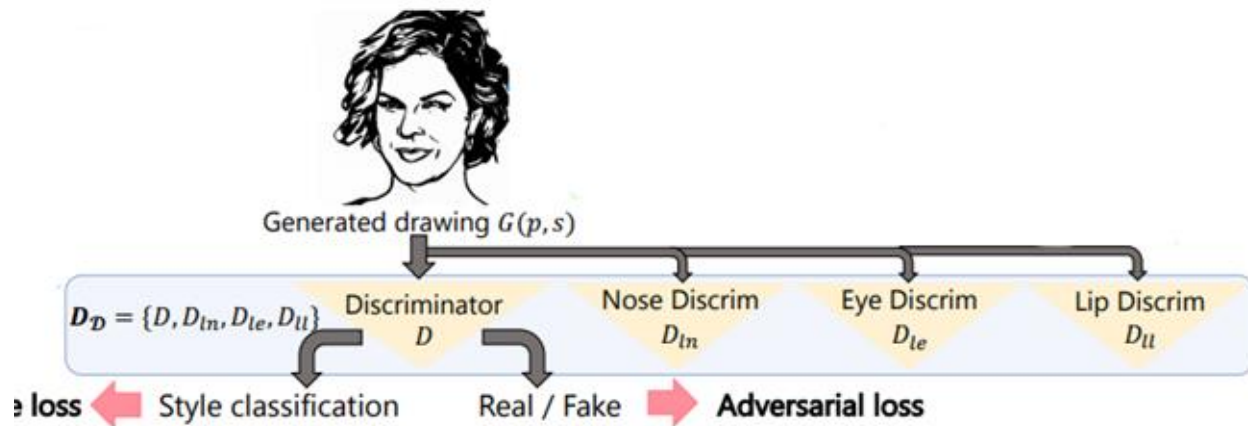
- We replicated the generator architecture in (Chan et al., 2022)'s work.
- We have 2 generators employing CycleGAN (Zhu et al., 2017) structure.

$$L_{cycle} = \|G_B(G_A(a)) - a\| + \|G_A(G_B(b)) - b\|$$

Figure Adapted from (Chan et al., 2022)

Adversarial loss: Discriminator

- We implemented UPDG discriminator (Yi et al., 2020) for nose, eye, and lips in addition to the global discriminator
- They all have PatchGAN (Isola et al., 2016) Discriminator architecture.
- Local discriminators receive masked input with mask obtained from (Gu et al., 2019) 's pretrained face parsing neural network.



Geometry Loss: SynergyNet (Wu et al. (2021))

- Encoding images into z vectors that carry information relating to that image's geometry

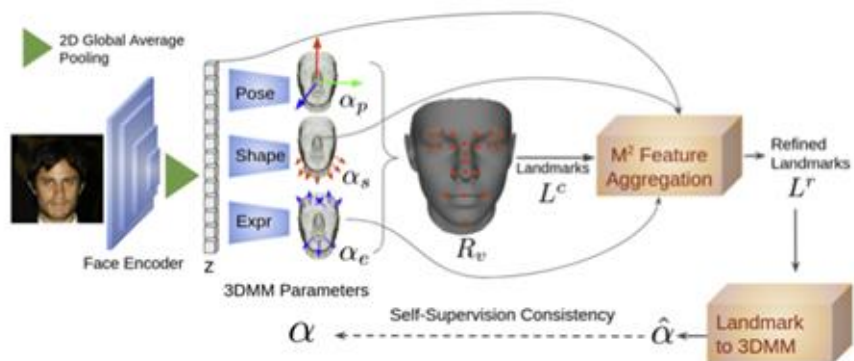


Figure Adapted from (Wu et al., 2021)



Figure Adapted from (Wu et al., 2021)

Semantic Loss: BiSeNet (Yu et. al (2018))

- Parsing network
- Pretrained on CelebAMask-HQ
- 19 classes including face components, hair, cloth and background
- Satisfactory results for both photo and sketch
- Loss is the distance between BiSeNet outputs of input photos and generated sketches

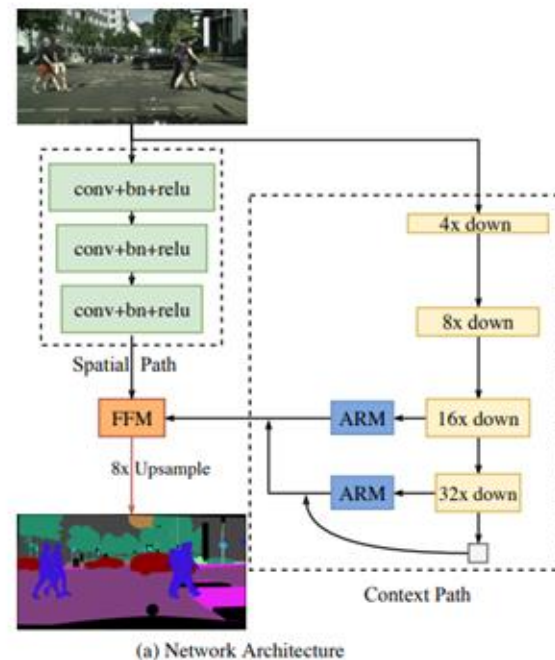
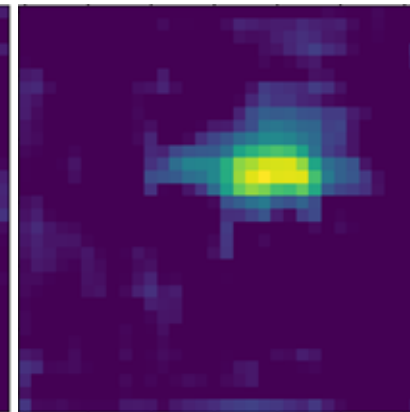
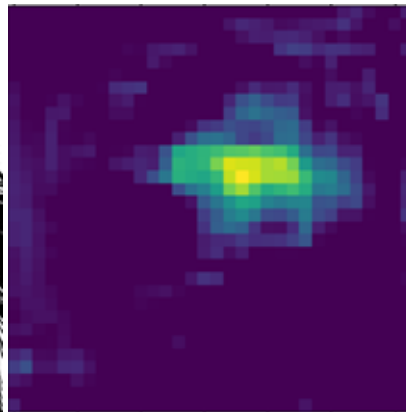


Figure: Model overview of BiSeNet
Adapted from (Yu et al, 2018)

BiSeNet Results

Here are our outputs for locations of left eyebrow for corresponding photo and sketch



CLIP Loss

- Following Chan et al. (2022), we used a CLIP embedding distance loss

$$L_{\text{CLIP}} = \|\text{CLIP}(G_A(a)) - \text{CLIP}(a)\|$$

Figure adapted from (Chan et al., 2022)

Training

Chan et al. (2022) trained their model on large datasets, which they could do because they worked on a general domain

Our training scheme:

Phase 1: Pre-training with samples from FFHQ (Karras et al., 2018) and Chan et al. (2022)'s model

Phase 2: Fine-tuning with human-drawn sketches from APDrawing Dataset (Yi et al., 2019)

Experimentation

- We did 9 experiments with different configurations and evaluated them according to our metrics to find which directions are more promising.
- Pretraining+Fine Tuning
- For pretraining, we follow two main types of training:
- Type 1: Adversarial Loss, CLIP Loss, Semantic Loss through Face Parsing Network (Gu et al., 2019)
- Type 2: The whole methodology

Experiments

1. Type 2 Training + Fine Tuning
2. Just Fine Tuning
3. Type 1 Pretraining
4. Type 2 Training
5. Type 2 Training - SynergyNet
6. Type 1 Pretraining + (Type 2 Training - SynergyNet)
7. Type 1 Pretraining + Type 2 Training
8. Type 2 Training with High Adversarial Loss
9. Type 1 Pretraining + (Type 2 Training with High Adversarial Loss)

Samples



Experiment 1



Experiment 2



Experiment 3



Experiment 4



Experiment 5



Experiment 6



Experiment 7



Experiment 8



Experiment 9



Baseline



Ground-truth
(for evaluation only)



Photo

Experimental Results

Model	FSIM ↑	FID ↓	Face recognition accuracy ↑
Baseline	0.58	5.18	0.5
Experiment 1	0.51	50.46	0.63
Experiment 2	0.54	65.50	0.67
Experiment 3	0.41	24.94	0.6
Experiment 4	0.59	42.13	0.6
Experiment 5	0.55	54.27	0.6
Experiment 6	0.34	113.12	0.53
Experiment 7	0.44	11.81	0.1
Experiment 8	NaN	171.46	0
Experiment 9	0.33	43.70	0

Datasets

- FFHQ Dataset (Karras et al., 2018) for training
- Sketches generated by applying (Chan et al., 2022)'s model on FFHQ as sketch dataset for training
- APDrawing Dataset (Yi et al., 2019) for fine tuning and evaluation

Metrics (following (Yu et al., 2020)'s work)

- **FID** (Heusel et al., 2017): evaluation of sketches in terms of realism and variation
- **FSIM** (Zhang et al., 2011): evaluation of similarity between generated sketches and ground truths
- **Face Recognition Accuracy**: evaluation of how much the identities are accurately represented in sketches via accuracy of a face recognition tool employing state-of-the-art models
(https://github.com/ageitgey/face_recognition)

Limitations

- Model quality depends on another model
- Training progress is very sensitive to loss weights. In some cases we saw that our model favoured generating grayscale versions of the images which we hypothesize is because the weights of feature based losses were too large and the model learned that it could optimize the overall loss best by not deviating from the original image much.

Future Work

- A larger dataset of face sketches could be useful as there is a lack of such data
- It can be trained with different styles of sketches separately

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